

M N Aditya

Moodur Narayanan Aditya

Chennai, Tamil Nadu, India

Mobile: +91 9500042517

Email: an8594@srmist.edu.in

LinkedIn: linkedin.com/in/aditya-narayanan-0820392a4

GitHub: github.com/MNADITYA05

Google Scholar: Scholar Profile

ResearchGate: researchgate.net/profile/M-N-Aditya-2

Personal Profile

Final-year ECE undergraduate (Data Science) with **13 peer-reviewed publications** across Scopus/IEEE/Springer-indexed venues and research stints at **5 national institutes** (IITs, NITs, IIIT). Specialises in federated learning, medical image segmentation, and agentic LLM systems for edge-constrained deployment.

Education

2023 – 2027: B.Tech., Electronics & Communication Engineering (Data Science)

SRM Institute of Science and Technology, Chennai

CGPA: 8.17 / 10.00

Experience

Aug 2026 – Present: Research Intern (Remote)

Indian Institute of Technology, Kharagpur

Supervisor: Dr. Ishita Maiti

- Developing a **GNN-driven multi-class segmentation pipeline** for **lung CT and chest X-ray scans**, leveraging graph-structured representations for pathological region delineation across multiple disease categories.
- Designing graph neural network architectures to model spatial dependencies between anatomical regions for robust cross-modal medical image segmentation.

Jun 2026 – Jul 2026: AI Development Intern (On-site)

Farmience AgroTech, Chennai

- Built a **Deep Q-Network** agent within a **Federated Learning** framework to optimise irrigation scheduling and nutrient delivery for precision agriculture.
- Developed a **computer vision pipeline** for real-time crop anomaly detection integrated into autonomous agricultural systems.

Dec 2024 – Feb 2026: Student Researcher (part-time, concurrent)

DST-ANRF (SERB) Funded Project, SRM IST, Chennai

Supervisor: Dr. Varadaraju Magesh

- Delivered a validated clinical prototype for real-time orthodontic debonding force measurement using **FSRs**, custom PCB, and **Arduino Mega 2560** — published in *Trends in Biomaterials & Artificial Organs* (2026). (**Grant No: CRG/2022/005834**)
- Bridged embedded sensing and clinical orthodontics for quantitative intraoral pain assessment, enabling data-driven debonding protocols.

Dec 2025 – Jan 2026: Research Intern – Business Intelligence & AI Strategy (On-site, concurrent)

SRM Technologies Private Limited, Chennai

- Conducted research and competitive analysis of **pharmaceutical BPO companies** based in Japan, evaluating market positioning and operational capabilities.
- Performed research on **Agentic AI** for tender management automation, identifying opportunities for document processing and workflow optimization.

May 2025 – Jul 2025: Research Intern (Remote)

National Institute of Technology, Kurukshetra

Supervisor: Dr. Ghanapriya Singh

- Developed **FedDQPrune**, a DQN-driven adaptive pruning framework for federated **soybean yield prediction** on a 1,499-sample dataset across 5 clients, achieving **RMSE 3.57 bu/acre** and **R² 0.87** — outperforming FedAvg, FedPruning, and 1-shot-LT baselines; extended into **FedDQN-Ctrl**. (*SiMPaCT 2026*)
- Reduced model size by **28.7%** (628K → 448K parameters) with **28.7% communication savings** via mask-aware federated aggregation for edge agricultural IoT deployment.

May 2025 – Jul 2025: Research Intern (Remote, concurrent)

Indian Institute of Technology, Jammu

Supervisor: Dr. Badri Narayan Subudhi

- Developed a **tri-modal forgery detection framework** (ResNet-50 visual + FFT frequency + HOG gradient) with **TriModalCross-Attention** fusion, achieving **69.56% test accuracy** and **AUC-ROC 0.7614** on the RTM dataset.
- Deployed a **3-head ensemble meta-classifier** with adaptive architecture scaling and EMA weight averaging, outperforming Generic CNN and ManTra-Net baselines.

Apr 2025 – Jul 2025: Industrial Trainee (concurrent)

Intel Corporation — Unnati Industrial Training Programme

- Built **TRACE**, an end-to-end IoT traceability pipeline using **ESP32 + GM805L** barcode scanner and **YOLOv11n** PCB defect detection across **7 defect classes (>95% precision, >85% mAP@0.5)**.
- Automated the scan-to-result pipeline — barcode verification, defect classification, and **MongoDB** audit logging — with real-time operator feedback via TFT display and a live **Streamlit** monitoring dashboard.

Dec 2024 – Jan 2025: Research Intern (Remote, concurrent)

National Institute of Technology, Tiruchirappalli

Supervisor: Dr. G.R. Gangadharan

- Designed **SMT-UPerNet** for liver tumour segmentation on **LiTS17**, achieving **89.00% Dice** and **80.20% IoU** — outperforming Swin-Unet by **+0.60% Dice** under identical conditions. (*HINT 2025, Scopus-indexed*)
- Integrated **Grad-CAM** into the segmentation decoder with **progressive layer-unfreezing** and **EMA** training, achieving **HD95 3.28 mm** boundary accuracy.

Dec 2024 – Jan 2025: Research Intern (Remote, concurrent)

Indian Institute of Information Technology, Allahabad

Supervisor: Dr. Ramesh Kumar Bhukya

- Fine-tuned **AI4Bharat IndicConformer** (129M params) for low-resource **Kashmiri ASR** using **NeMo** with **Hybrid CTC/RNNT** decoding, reducing WER from **39.1%** (IndicVoices baseline) to **14%** over 50 epochs.
- Built dynamic audio chunking, regex-based Kashmiri text normalisation, and a custom **WER callback** for real-time training monitoring across a 5,632-token multilingual vocabulary.

Publications

Journal Articles

Development of a Device Prototype for Orthodontic Debonding Force Measurement. *Trends in Biomaterials and Artificial Organs*, 2026. [Link]

Explainable AI for Panoramic Dental Radiographs using Contrastive Learning and UNet Based Segmentation. *Journal of Soft Computing Paradigm*, Vol. 7, No. 2, pp. 114–123, 2025. [Link]

Conference Papers

FedDQN-Ctrl: Dynamic Client Compression for Federated Learning. **SiMPaCT 2026**. [Link forthcoming]

TinyFedPrompt: Federated Prompt Tuning of BioMedCLIP for Privacy-Preserving Pneumonia Detection on Edge Hardware. **PACCAS-2026**. [Link forthcoming]

An FPGA-Based Architecture for Interpretable PM2.5 Anomaly Detection. **ICIET-2026**. [Link forthcoming]

Deep Learning for Inflation and Oil Price Forecasting: A Comparative Study of LSTM and Transformer Models for Macroeconomic Policy Analysis. *Innovations in Data Analytics: Selected Papers of ICIDA 2025, Volume 1*, Springer LNNS (Scopus-indexed), pp. 339–361. [Link]

SMT-UPerNet: A Scale-Aware Modulation Transformer Network for Explainable Liver Tumor Segmentation. **HINT 2025**, *Springer LNNS Proceedings* (Scopus-indexed). [Link]

Few-Shot Learning for Early Detection of Laryngeal Cancer with Explainable AI. **IEMENTech 2025** (IEEE). [Link]

Explainable AI for Kidney Stone Prediction: A Machine Learning-Based Clinical Analysis. **RAEEUCCI 2025** (Scopus/IEEE). [Link]

Malware Analysis Using Hashing and Explainable AI: A Comparative Study of LIME and SHAP Techniques. **AIML-AE 2025** (IEEE). [Link]

Analyzing Land Submergence Impact on Temperature Using Random Forest Regression and Explainable AI. **ACOIT 2024** (Scopus/IEEE). [Link]

Book Chapters

Converging Sustainability, AI and Global Health: A Data-Driven Roadmap for Responsible Trace Element and Heavy Metal Management. *Trace Elements and Heavy Metals: Bridging Sustainability and Global Health Goals*, Springer, pp. 565–599, 2026. [Link]

AI-Driven Circular Economy: Optimizing Industrial Use and Recovery of Trace Elements and Heavy Metals. *Trace Elements and Heavy Metals: Bridging Sustainability and Global Health Goals*, Springer, pp. 535–564, 2026. [Link]

Projects

Loan Default Prediction Dashboard | 2026 [GitHub]

- Achieved **ROC AUC 0.84** on **150K real borrowers** by training a **Random Forest** classifier with **SHAP** per-prediction waterfall explanations for interpretable credit risk assessment.
- Enabled real-time portfolio monitoring by logging predictions to a **SQLite** backend and connecting a live **Power BI** reporting layer via DirectQuery.

DealSight – M&A Deal Success Predictor | 2026 [GitHub]

- Achieved **CV ROC-AUC 0.78** on M&A deal outcome prediction by training an **XGBoost** pipeline on 18 financial and market features ingested from **SEC EDGAR** filings and **yfinance** market data.
- Delivered per-prediction explainability by integrating **SHAP** waterfall analysis, served via a **FastAPI** backend and **Streamlit** UI, containerised with **Docker** and CI via **GitHub Actions**.

AI-Enhanced Financial Distress Prediction for SMEs | 2026 [GitHub]

- Delivered explainable SME risk assessments by building a **LangGraph multi-agent pipeline** powered by **Groq LLaMA 3.1** with an **ensemble of 4 ONNX-optimised models** (DNN, RF, LR, GB).
- Enabled grounded financial advisory by deploying a **RAG chatbot** with **Gemini embeddings** and **MongoDB Atlas Vector Search** via **FastAPI** and **Next.js**.

TinyFedPrompt – Medical Image Classification | 2026 [GitHub]

- Achieved **AUC 0.9509** (X-ray) and **AUC 0.9419** (CT) by developing a federated pipeline with prompt-tuned **BioMedCLIP ViT** for privacy-preserving medical image classification.
- Enabled resource-constrained edge deployment by exporting the model via **ONNX** and running inference on **Raspberry Pi 4B**.

DentoSeg-SSL: Self-Supervised Dental Panoramic Segmentation | 2025 [GitHub]

- Surpassed **state-of-the-art** segmentation performance with a **92.7% Dice coefficient** by designing a **U-Net** with **contrastive learning** pre-training on dental panoramic radiographs.
- Enhanced clinical interpretability by integrating **Grad-CAM** visualizations for explainable tooth and tissue segmentation predictions.

Skills

Languages: Python, C, MySQL

Frameworks: PyTorch, TensorFlow, Hugging Face, NeMo, FastAPI, Streamlit, Next.js, n8n

ML & AI: Federated Learning, Reinforcement Learning, LangGraph, RAG, ONNX, SHAP, XAI

Computer Vision: YOLO, OpenCV, Grad-CAM, Albumentations

Embedded & Hardware: ESP32, Arduino, FPGA, PlatformIO, I2S

Databases: MongoDB Atlas, SQLite

Dev Tools: Git, Docker, GitHub Actions, MLflow, Jupyter, Power BI

Achievements

- 2nd Place** (Rs. 12,000) – Code Merge Hackathon (24-hour, 150 participants, Taiwan-India Joint Living Lab on Big Data Analytics & DSBS, SRMIST) 2025
- Best Paper – 2nd Prize** (Rs. 7,500) – RISXAI 2024, S.A. Engineering College (IEEE CS Madras, CSI & ACM Chennai; 14 papers) 2024
- Finalist** – SAMHITA '24 Hackathon (National level, MIT Anna University Chennai, 170 participants) 2024

Invited Talks & Workshops

- Workshop on Code to Core — C Programming 2026
- Workshop on Introduction to Computer Vision and its Application in Agriculture at Chettinad – Sarvalokaa Education, International School 2025
- Selected for 8th Summer School on AI & 3DVSS organized by **IIIT Hyderabad** 2025
- Guest Speaker on Human-Robot Interaction at **IEI, KTR Local Center** 2024